Capstone Project 5: Serverless "Event-Driven" Image Processing Pipeline

Scenario: A digital asset management company needs to automate the generation of thumbnails and watermarks for millions of high-resolution images uploaded daily.

- **Problem Statement:** Manual processing is too slow. Using dedicated servers for 24/7 processing is expensive and inefficient for bursty workloads.
- **Case:** Implement a 100% serverless workflow that triggers instantly on upload, uses "Dead Letter Queues" (DLQ) for error handling, and secures assets at rest.
- **Solution Components:**
  - **Storage:** S3 (Deep Dive: Lifecycle policies, Versioning).
  - **Compute:** Lambda (Triggered by S3).
  - **Workflow:** SNS for fan-out notifications; SQS as a buffer for heavy processing.
  - **Secret Management:** SSM Parameter Store for API keys.

Terraform Snippet (S3 to Lambda Trigger)

Terraform
```
# S3 Bucket with Versioning & Encryption
resource "aws_s3_bucket" "uploads" {
  bucket = "ssdn-asset-uploads"
}

# Lambda Trigger Configuration
resource "aws_s3_bucket_notification"
"bucket_notification" {
  bucket = aws_s3_bucket.uploads.id

  lambda_function {
    lambda_function_arn =
aws_lambda_function.processor.arn
    events              = ["s3:ObjectCreated:*"]
  }
}

# Secure Secret in SSM
resource "aws_ssm_parameter" "watermark_api_key" {
  name  = "/prod/assets/watermark_key"
  type  = "SecureString"
  value = var.secret_api_key
}
```